

IC 111: Linear Algebra

Course Code: IC 111

Course Name: Linear Algebra

L-T-P-C: 2.5-0.5-0-3

Prerequisite: Consent of the faculty member

Students intended for: B.Tech 1st year

Elective or Core: Core

Approval: 2nd Senate

Course contents

- **Matrix Theory:** Rank of Matrix, inverse of a matrix by elementary operations, Solution of linear simultaneous equations and their numerical solutions by gauss Elimination and Gauss Seidel Methods. Eigen values and eigen vectors, Cayley Hamilton Theorem, Diagonalization of Matrices. Orthogonal, Hermitian, Skew Hermitian, Normal and Unitary matrices and their elementary properties, Quadratic Forms. [12 Lectures]
- **Vector Spaces:** Vector spaces, Sub Spaces, Linear Dependences and Independences of Vectors, Span, Bases and Dimensions, Direct Sum. [12 Lectures]
- **Linear Transformations:** Linear Transformations, Linear Variety, Range Space and Rank, Null Space and Nullity, Homomorphism, Matrix of Linear Transformations, Matrix Representation of a linear transformation, Structure of the solutions of the matrix equation $Ax = b$, Change of bases. [12 Lectures]

Text Books:

1. G.Strang, **Linear Algebra and its Applications**, 4th Edition, Thomson, (2006).
2. K. Hoffman and R. Kunze, **Linear Algebra**, Prentice Hall, (2008).
3. H.Anton, **Elementary Linear Algebra with Applications**, 9th Edition, John Wiley (2004).

References:

1. E.Kreyszig, **Advanced Engineering Mathematics**, 9th Edition, John Wiley (2007).
2. S.Kumaresan, **Linear Algebra – A Geometric Approach**, Prentice Hall of India (2004).
3. D. S. Watkins, **Fundamentals of Matrix Computations**, 2nd Edition, John Wiley & Sons (2002).